

LegalEase

Phase 4: Project Planning

AI-Powered Legal Document Drafting and Export Platform

Project Phase Document

This PDF consolidates the prepared project materials for this phase.

README

Phase 4 — Project Planning

4.1 Development Plan

Stage	Work	Deliverable
1	Repository setup	Project skeleton
2	Environment setup	.env, requirements
3	Backend	FastAPI application
4	AI layer	Gemini generator
5	Frontend	Streamlit application
6	Export	TXT/DOCX/PDF
7	Testing	Unit/API/export tests
8	Documentation	User/developer docs
9	Demo	Presentation and walkthrough

4.2 Milestones

M1 — Foundation

Repository, virtual environment, dependencies, configuration.

M2 — API

Schemas, routes, health endpoint, generation endpoint.

M3 — AI

Gemini integration and prompt template.

M4 — UI

Form, loading state, preview, download controls.

M5 — Export

TXT, DOCX, PDF generation.

M6 — Quality

Validation, exceptions, tests.

M7 — Submission

Documentation, screenshots, demo script.

4.3 Risks

Risk	Impact	Mitigation
Gemini API failure	High	Catch errors and allow retry

Risk	Impact	Mitigation
Invalid input	Medium	Pydantic + frontend validation
Hallucinated content	High	Controlled prompt + review warning
API key exposure	High	Environment variables
PDF formatting issue	Medium	Export tests
Dependency conflict	Medium	Pin/test dependencies
Large prompts	Medium	Input limits and clear validation

4.4 Definition of Done

A feature is complete when it is implemented, tested, documented, and demonstrated successfully.

02 Task Checklist

Implementation Checklist

Setup

- ☐ Create virtual environment
- ☐ Install requirements
- ☐ Create .env
- ☐ Verify Gemini key
- ☐ Start backend
- ☐ Start frontend

Backend

- ☐ backend/main.py
- ☐ backend/routes.py
- ☐ backend/schemas.py
- ☐ backend/ai_core/gemini_generator.py
- ☐ Health route
- ☐ Generate route
- ☐ Error handling

Frontend

- ☐ Form
- ☐ Validation
- ☐ API call
- ☐ Loading indicator
- ☐ Preview
- ☐ Download buttons

Testing

- ☐ Health API
- ☐ Valid generation request
- ☐ Missing field
- ☐ Empty terms
- ☐ AI failure
- ☐ TXT export
- ☐ DOCX export
- ☐ PDF export

Final

- ☐ README

- ☐ Screenshots
- ☐ Demo video
- ☐ Presentation
- ☐ Final repository cleanup

INDEX

Phase-4-Project-Planning

Files in this phase:

- 02-Task-Checklist.md
- README.md